

Iteration 0 - Proposal

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1. **Problem Statement:** Clearly describe what your software aims to address.
 - a. College students manage multiple courses with overlapping deadlines and often fail to distribute study time effectively, leading to last minute cramming. This project will build a Smart Study Planner with a graphical interface that helps students organize coursework and generate a weekly study schedule.
2. **Scope of your project:** Clearly specify what the software will and will not do.
 - a. **In Scope (What the software WILL do)**
 - i. Provide a GUI for managing courses and coursework (assignments/exams/projects).
 - ii. Allow users to enter: title, course, due date/time, estimated hours per course/credit hours, priority, and optional notes.
 - iii. Generate a suggested study schedule for the next 7 days (or user selected range) by allocating study blocks across days.
 - iv. Prevent scheduling past due items and clearly overdue tasks.
 - v. Allow marking tasks as “in progress” or “completed,” and update the schedule accordingly.
 - vi. Save and load data locally and remotely so that it can be installed on a University computer.
 - b. **Out Of Scope (What the software WILL NOT do)**
 - i. No integration with Canvas/Blackboard/Google Calendar.
 - ii. No accounts, cloud sync, or multi-device login.

- iii. No push notifications, email, or SMS reminders.
- iv. No “AI” features or learning from behavior.
- v. No group collaboration or shared schedules.

3. **Functional Requirements:** List the functional requirements within the specified scope as a list.

- a. **FR1:** The system shall allow users to create, view, edit, and delete a local user profile containing name, weekly study goal (hours/week), and preferred study block length (30/60/90 minutes).
- b. **FR2:** The system shall allow users to create, edit, and delete courses with course name, credit hours (1–6), and optional color/tag.
- c. **FR3:** The system shall allow users to create, edit, and delete tasks associated with a course. Each task shall include: task title, task type (assignment/quiz/exam/project), due date/time, estimated study hours (0.5–50), priority (Low/Medium/High), and optional notes.
- d. **FR4:** The system shall allow the user to set study availability per day of week in hours (0–12 hours per day).
- e. **FR5:** The system shall display all tasks in a list view and allow sorting by due date, priority, course, and completion status.
- f. **FR6:** The system shall allow users to mark tasks as Not Started, In Progress, or Completed.
- g. **FR7:** The system shall generate a study plan for a user-selected date range (default: next 7 days) by allocating study blocks across days based on due date/time, priority, and remaining estimated hours.

- h. **FR8:** The system shall not schedule any study block after a task's due date/time and shall flag tasks that cannot be fully scheduled before the due date/time.
- i. **FR9:** The system shall regenerate the study plan on demand and, when a task is marked Completed, remove any remaining scheduled blocks for that task from the plan..
- j. **FR10:** The system shall save and load all profile data, courses, tasks, and study availability to/from a local file so the application state persists across restarts.
- k. **FR11:** When creating a new task, the system shall auto-suggest estimated study hours using the course's credit hours as:
 - i. $\text{Suggested_hours} = \max(0.5, \text{credit_hours} \times 1.0)$. The user shall be able to accept the suggested value or override it.

4. **Non-functional Requirements:** List any constraints that the system will have.

- a. **NFR1:** The system shall store and load all user data locally and shall not require an internet connection for any functionality.
- b. **NFR2:** The system shall not collect, transmit, or share any user data over a network.
- c. **NFR3:** The system shall launch to the main interface within 5 seconds on the lab SoC computer.
- d. **NFR4:** The graphical user interface shall be available in English only.
- e. **NFR5:** The system shall generate a 7-day study plan for up to 200 tasks within 2 seconds on the lab SoC computer.

- f. **NFR6:** The system shall validate all required input fields and prevent saving data with missing or invalid values (e.g., invalid dates, negative hours, credit hours outside 1–6).